

Design of Generative Multimodal AI Agents to enable Persons with Learning Disability

Rajagopal A*
Indian Institute of Technology,
Madras, India
rajagopal.motivate@gmail.com

Nirmala V
Queen Mary's College, Chennai, India
gvan.nirmala@gmail.com

Immanuel Johnraja Jebadurai
Karunya Institute of Technology &
Sciences, Coimbatore, India
immanueljohnraja@karunya.edu

Arun Muthuraj Vedamanickam
National Institute of Technology,
Trichy, India
arun.gvan@gmail.com

Prajakta Uthaya Kumar
National Institute of Design,
Bengaluru, India
prajakta.uk490@gmail.com

ABSTRACT

The recent advances in Multimodal AI & Generative AI open doors to the possibilities of solving key challenges for Persons with Learning Disability. To assist individuals facing difficulty in visual or auditory perception, this paper designs & develops a multimodal AI agent using recent advances in the field. We aim to solve the challenge of enabling persons with Visual or Auditory Processing Disorders to learn & communicate. We do this by exploring a design that allows the transformation of information across visual and language modalities. This design can be realized with the recent advances in Generative Multimodal AI. Based on each individual's needs, the AI agent dynamically adapts the Human Computer interaction model. For instance, for a child with Visual Processing Disorder (VPD), given the child's hindered ability to make sense of information taken in through the eyes, the Multimodal AI agent transforms any visual information into auditory user interaction. In another instance, for a person with Central Auditory Processing Disorder (CAPD), given the hindrance in the individual's ability to analyze information taken in through the ears, the AI dynamically translates any speech modality into visual cues. Thus the AI agent adapts dynamically to the strengths and abilities of the individual. To enable students with VPD to learn, the design allows the student to ask questions about an image. This design is realized as a Visual Question Answering task in Vision Language Transformer models. We explore interactive multimodal conversations with Few shot Learning and In-Context Instruction Tuning of Multimodal Large Language Models to address difficulty in visual reasoning. To enable persons with CAPD to learn, the design translates audio lectures into visual cues. This visual cue consists of a combination of words using speech recognition and Large Language Models based re-phrasing to simpler words, cross-modal retrieval of images to address auditory memory challenges, and AI-generated images. To

identify the strengths of each child, we also explore Multimodal embedding based Multimodal latent space arithmetic to link AI across senses. To effectively integrate the proposed design into the mainstream, we explore a universal design based inclusive approach to extend the use case to create AI assistants for assisting children with different learning styles such as visual learners or auditory learners. To enable future research on the proposed design, we explore an architecture to compose a pipeline of AI models, and to connect with external systems via plugin connectors. We implement lab scale prototypes of this design and present a demo on the project webpage at <https://sites.google.com/view/multimodallearningdisability>.

CCS CONCEPTS

• Artificial intelligence; • People with disabilities; • Accessibility; • Human computer interaction;

KEYWORDS

Generative AI, Multimodal AI, Human Computer Interaction, Assistive Technology, Learning disability, Person with Disability, Central Auditory Processing Disorder, Visual Processing Disorder, Multimodal Large Language Transformers, Multimodal Few shot Learning, Multimodal In-context Tuning, Multimodal latent space, Vision Language Models

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1 INTRODUCTION

1.1 Motivation to create Multimodal Assistive Technology to enable People with Learning disorders

Neurodevelopmental disorders affect the learning ability of an individual. Learning disabilities are neurobiological factors that alter brain functioning in a manner which affects cognitive processes related to learning. These processing problems can interfere with learning basic skills. Among students who received special education under the Individuals with Disabilities Education Act, the

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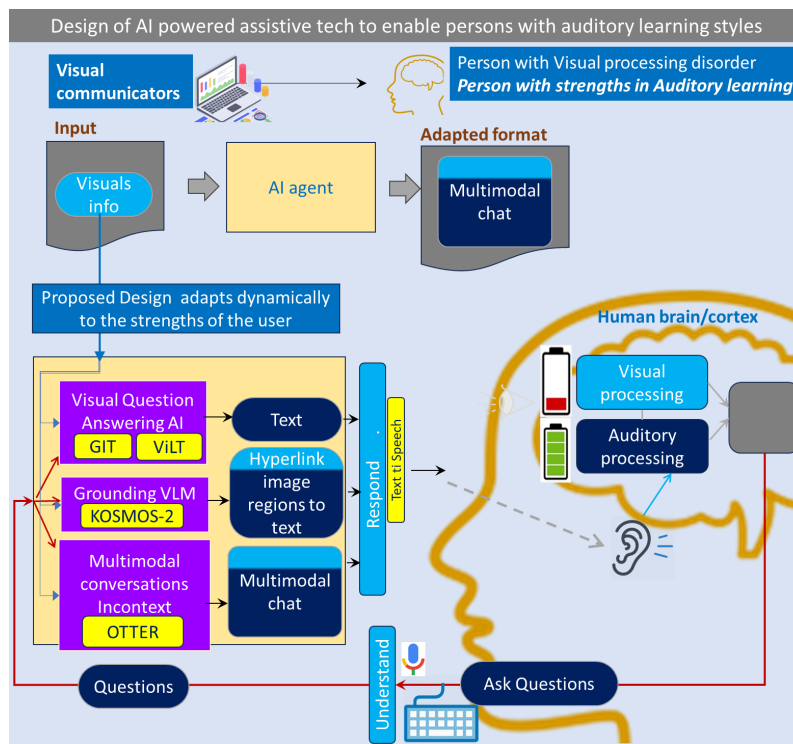


Figure 1: Design of MAA to enable a person with VPD

largest reported percentages of students were specific learning disabilities as per recent statistics by the National Center for Education Statistics [1]. A learning disability is a lifelong challenge. This paper focuses on two common areas of difficulty associated with learning disabilities (LD) namely visual and auditory perception [2]. A hindered ability to make sense of information taken in through the eyes is Visual processing disorder (VPD) [2]. Difficulties with visual processing affect how visual information is processed by the brain. A hindered ability to analyze or make sense of information taken in through the ears is Auditory processing disorder (APD) [2]. Difficulties with auditory processing affect how this information is interpreted by the brain. However, with appropriate intervention, people with learning disabilities can achieve success at school, workplace, relationships, community [3]. As per the Learning Disability Association of America, people with learning disabilities are of average or above average intelligence [3]. Hence, the motivation for research and development of interventions for enabling persons with Learning Disability. Given the recent advances in the field of AI, the opportunities to contribute is enormous. The advances in Multimodal Large Language models [7] or Multimodal Generative AI [6-15] have opened the doors to contribute to the conceptualization of innovative assistive technology to enable persons with Neurodevelopmental disorders. Hence, it is important to explore the possibilities of designing assistive tech for persons with learning disorders in the context of these recent advancements in Multimodal Generative AI, Vision Language Foundation models [6].

Millions of children with learning and attention issues have not received specialized accommodation as per the National Center for Learning Disabilities, in US [4]. In the US, 1 in 5 children have learning and attention issues [4]. National Center for Learning Disabilities identifies the barriers to success, which include lack of identification and effective intervention [4]. The ability to learn and succeed in school & work, and in relationships often depends on information exchanges across multiple human senses. Given the human potential, and the opportunities enabled by disability laws [16], an effective design of solutions is essential. Such solutions may tap into the strengths of each child. We discuss approaches for two types, VPD & APD in the next two paragraphs.

1.2 How to enable persons with Visual Processing Disorder?

Since so much information in the classroom and workplace is presented visually, persons with Visual Processing Disorder (VPD) can be at a disadvantage. When a person experiences hindered ability to make sense of information taken in through the eyes or experiences difficulty in visual perception, how to enable them to handle visual information with their strengths? Since each child has unique strengths [4], the AI should dynamically adapt the information to suit the needs of each child. When the student is seeing diagrams in an e-learning classroom or when a co-worker presents a visual Powerpoint in an office meeting, how does the person with VPD understand this information? If the AI can identify the strengths of

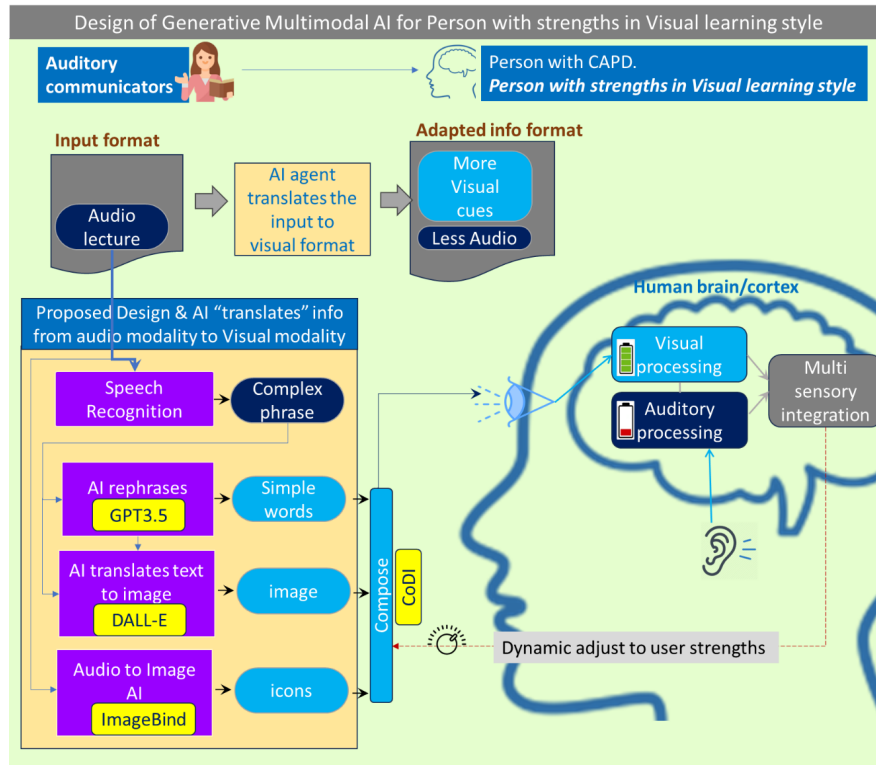


Figure 2: Design of MAA to enable a person with CAPD

each individual such as strengths in Auditory learning styles or Non-Visual interpretation, imagine a multimodal translator [17] that can transform information across visual and auditory modalities. Such Multimodal Agents can offer effective assistance when the person's brain has difficulty processing information in one of the modalities. Fortunately, the recent advances in the field of AI have opened up the possibility of development of the proposed Multimodal AI powered Assistive agents [16]. The availability of Pretrained Vision Language models [5, 6] and Multimodal Foundation Models [7] has opened up the opportunity for research and development of proposed AI agents to enable persons with difficulty in visual perception. There is an opportunity to contribute to research & develop assistive tech that adapts information for persons with VPD. The challenges faced by these individuals can be addressed by mapping them into problem statements in the AI domain. The challenges can be addressed by creative use of Vision-Language modeling (joint learning of visual and text info) such as ViLT/GIT [5, 6], Multimodal Large Language models [7], Visual Question Answering (VQA) tasks [5], Multimodal In-context [7], Generative AI models that ground generated text on visuals (KOSMOS) [11]. To simplify, the idea is to tap into patterns of relationship between visual and textual modality learned by a large scale Deep neural network, and hence use these learnings to either translate across modalities or reason across both visual and language information. In short, the idea is to instruct a large scale Deep learning Foundation model to do any requested tasks such as multimodal reasoning, visual question answering, and translation between visual and text

modalities. The idea is articulated in Fig 1. As seen in this figure, the proposed solution enables persons with VPD by providing an accommodation, where complex visual input from the environment is transformed into a spoken human computer interaction model by clever use of a combination of Vision Language Models (ViLT) and Multimodal Generative AI models (OTTER).

Research gap/questions: The research gap is the application of these recent AI models [6, 7, 11] in the design and development of Multimodal Assistive Technology to enable persons with VPD. The recent & fast changing progress in the field opens up the opportunity to explore the application of Multimodal Generative AI in the development of personalized assistive tech for persons with learning disability. Immense possibilities has opened up starting June 2023 for Multi-modal In-context Tuning [7]. This can solve many challenges for persons with different types of learning disorders. Persons with LD often require assistance in new types of multimodal tasks such as visual discrimination, visual closure, visual agnosia [2]. In-Context Tuning enables to tune of Multimodal Foundational AI model to perform new types of multimodal tasks [18] with just a handful of training data. The research gap is to employ Multimodal In-Context Tuning [7] to the specific requirements of different persons with VPD.

The research question explores how to employ In-context Tuning [7] and Few Shot Learning [8] to develop Multimodal Assistive Agents to support persons with learning disability? How to design & develop Multimodal Assistive Agents for persons with difficulty in learning. How to employ Multimodal translation to transform

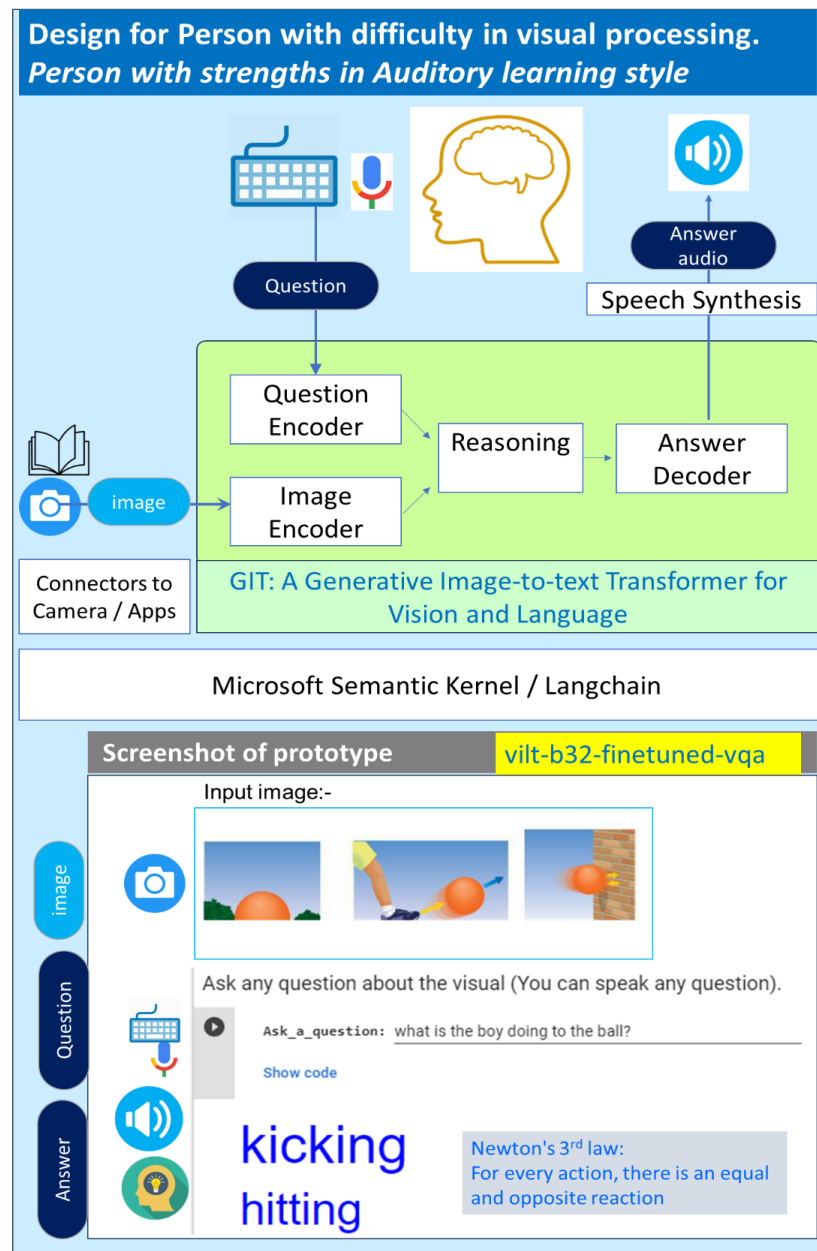


Figure 3: Prototype of interacting with an image by verbal dialog

information from one modality into another modality? How to enable persons with VPD to comprehend complex visuals by asking questions about the images, and hearing the answers with Visual Question Answering. How to design Multimodal Conversational agents by applying Multimodal Foundation models. How to develop prototypes of Multimodal assistive AI agents (MAA)?

1.3 How to enable persons with Central Auditory Processing Disorder?

With a hindered ability to analyze information taken in through the ears, a person with Auditory Processing Disorder (APD) has difficulty understanding oral directions or oral lectures. When instruction in school relies primarily on spoken language, the individual with an auditory processing disorder may have serious difficulty understanding the lesson. Similarly, when an office supervisor gives verbal instructions on a phone call, how does the person with APD understand & respond?. How to design an agent

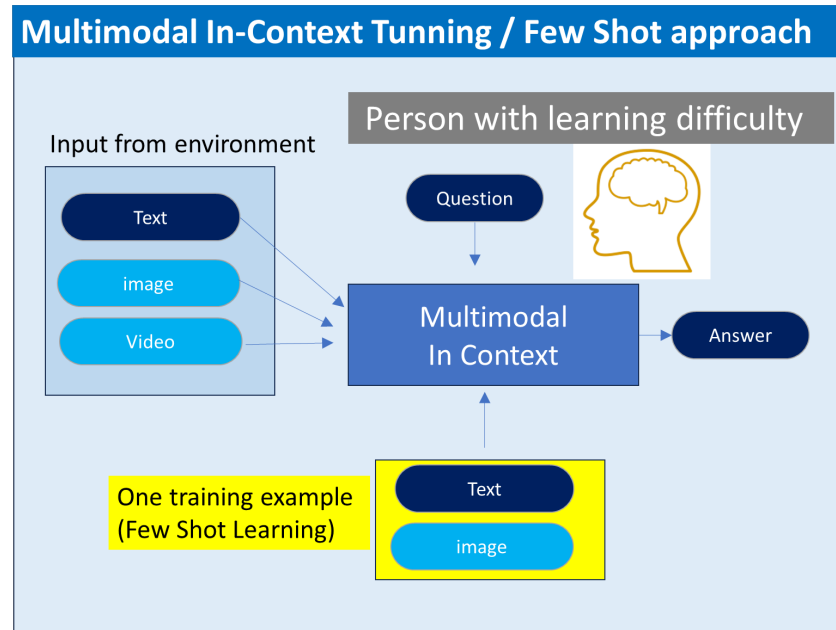


Figure 4: Potential to solve LD challenges by applying Multimodal InContext / Few shot learning

that dynamically adapts to the strengths of the individual?. The terms Auditory Processing Disorder (APD) or Central Auditory Processing Disorder (CAPD) are used interchangeably in this paper. The idea is illustrated in Fig 2. As seen in this figure, the proposed solution enables persons with CAPD by offering an accommodation, where any noisy/complex audio instruction is transformed into visual cues & easy to understand phrases by clever use of Large Language Models, and Generative Image generation AI models.

Research gap/questions: Given the enormous industry adoption of Generative AI during the last 6 months [14, 15], the opportunity is opening up to apply these advances in Multimodal Generative AI to solve the challenge of building personalized assistive tech for people with CAPD. Since 2023, the availability of advanced OpenAI models (GPT4, DALE-E) as REST APIs has practically opened doors for deploying smartphone based Assistive Technology for Persons with Learning Disabilities. With the mainstream availability of gigantic multimodal Foundation models for Inference tasks on HTTP end-points, such powerful AI can be configured and deployed to assist Persons with LD. Given such advanced Generative AI apps can now be deployed in smartphones [19], the possibilities for innovating Assistive Technology (AT) solutions for persons with learning disorders is suddenly expanding. It is important to explore this opportunity to conceptualize Multimodal Assistive AI agents (MAA) for persons with learning difficulties. Hence this area of creating personalized AT powered by Multimodal Generative AI can now not only be researched, but also deployed on millions of smartphones [15, 19] to transform the lives of persons with learning disability. This presents a significant opportunity for the researchers. When a person is experiencing difficulty in understanding spoken inputs, how to employ a smartphone MAA

that assists the person in understanding the inputs. Can the spoken inputs be dynamically translated into a visual medium?. Given the challenges in challenges of persons with CAPD to follow verbally spoken complex multi-step directions [2], how to use AI to rephrase complex oral phrases into simpler sentences. Given the memory challenges in remembering a verbal story [2], how can AI to dynamically convert the story into visual image sequences?

1.4 Contributions

The Design & Development of Multimodal Assistive Generative AI Agents (MAA) for persons with Learning Disability using the latest research advancements in AI is the contribution of this paper. Specifically, we aim to apply recent advances in Multimodal/Generative AI to enable persons experiencing difficulty in visual perception or auditory perception. By creatively mapping the challenges of persons with Visual Processing Disorder (VPD) or Auditory Processing Disorder (APD) into problem statements in AI, we design solutions. By identifying the individual's unique preferences in either visual learning style or auditory learning style, MAA dynamically provides appropriate accommodations to address the challenges. The design & development of MAA for persons with difficulty in Visual Perception & Auditory Perception is contributed in this paper. MAA has broader applications too, for example, a visual learner may employ MAA to easily learn from audio lectures, and vice versa.

The paper attempts to address the Learning Disability challenges by framing interventions as Multimodal AI challenges.

- **Enabling persons with VPD:** For Persons with strengths in Auditory learning style and difficulty in visual perception, we design the challenges into tasks in Multimodal/Generative

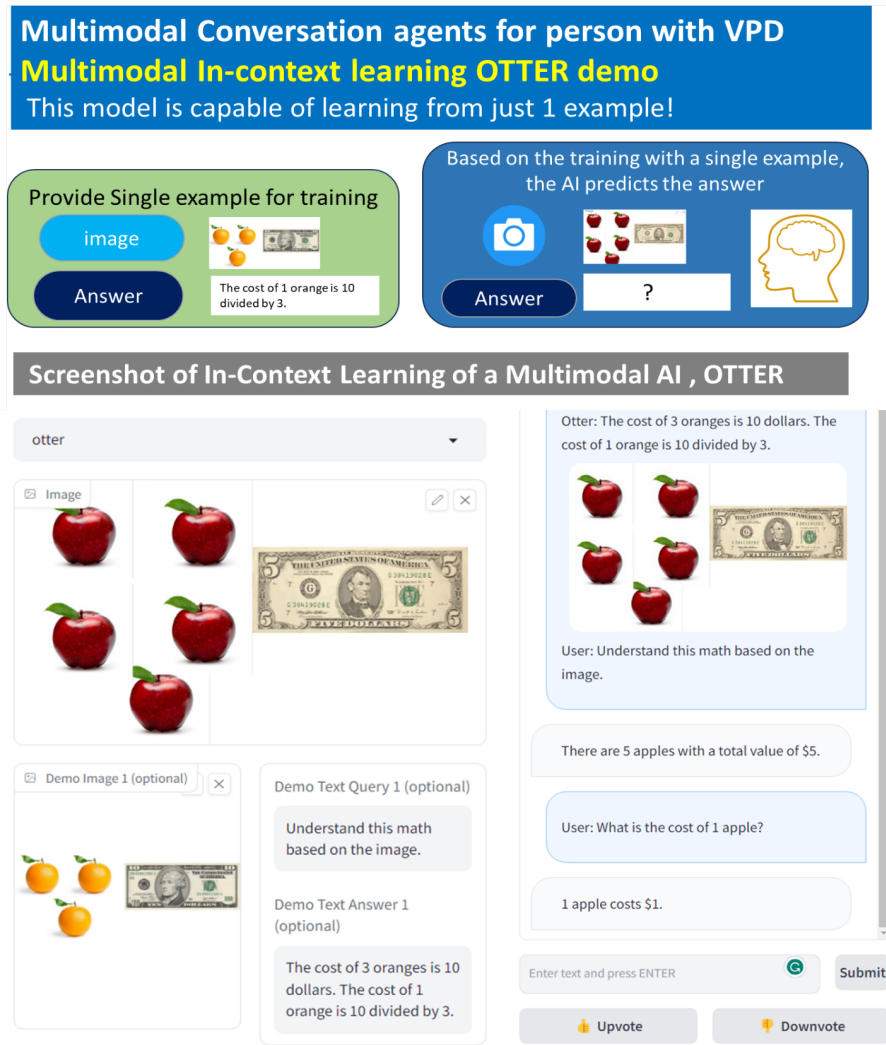


Figure 5: Multimodal Conversations AI agent with One-shot learning on a new math task

AI & demonstrate the solution with a lab scale demo on the project website.

- **Verbal Interaction with Images:** When the child faces difficulty in interpreting complex visual inputs [2], how to translate the information to a modality that is easily absorbed by the child? When the child experiences visual discrimination difficulties [2] affecting the ability to gain info from visuals, we explore the use of Vision Language Foundation models to allow the child to ask questions about an image, and then hear an answer. (Fig 3).
- **Reasoning over multiple multimodal content:** When the individual has difficulty in visual memory/ visual reasoning [2], how to assist the individual ? We explore approaches to solve this challenge by tapping into Multimodal Large models with In-Context Tuning on new customizable tasks. (Fig 4, Fig 5)

- **Enabling persons with CAPD:** Similarly, for Persons with strengths in Visual learning style and difficulty in auditory perception, the paper maps the challenges into tasks in Generative AI, and demonstrates a lab scale demo on the project website.
- **Multimodal Translation task:** When the person faces difficulty understanding orally spoken directions due to Auditory memory difficulty [2], how to employ AI to provide accommodation?. (Fig 2). We explore the feasibility of Multimodal Generative AI be employed to MAA translate complex spoken phrases into a combination of visual cues & simpler phrases? (Fig 6).
- **Multimodal embedding:** We explore the use of Multimodal latent space to link across modalities. (Fig 7)

By mapping the challenges faced by persons with learning disability into tasks in the domain of Multimodal/Generative AI, the paper designed a solution to enable persons with difficulty in either

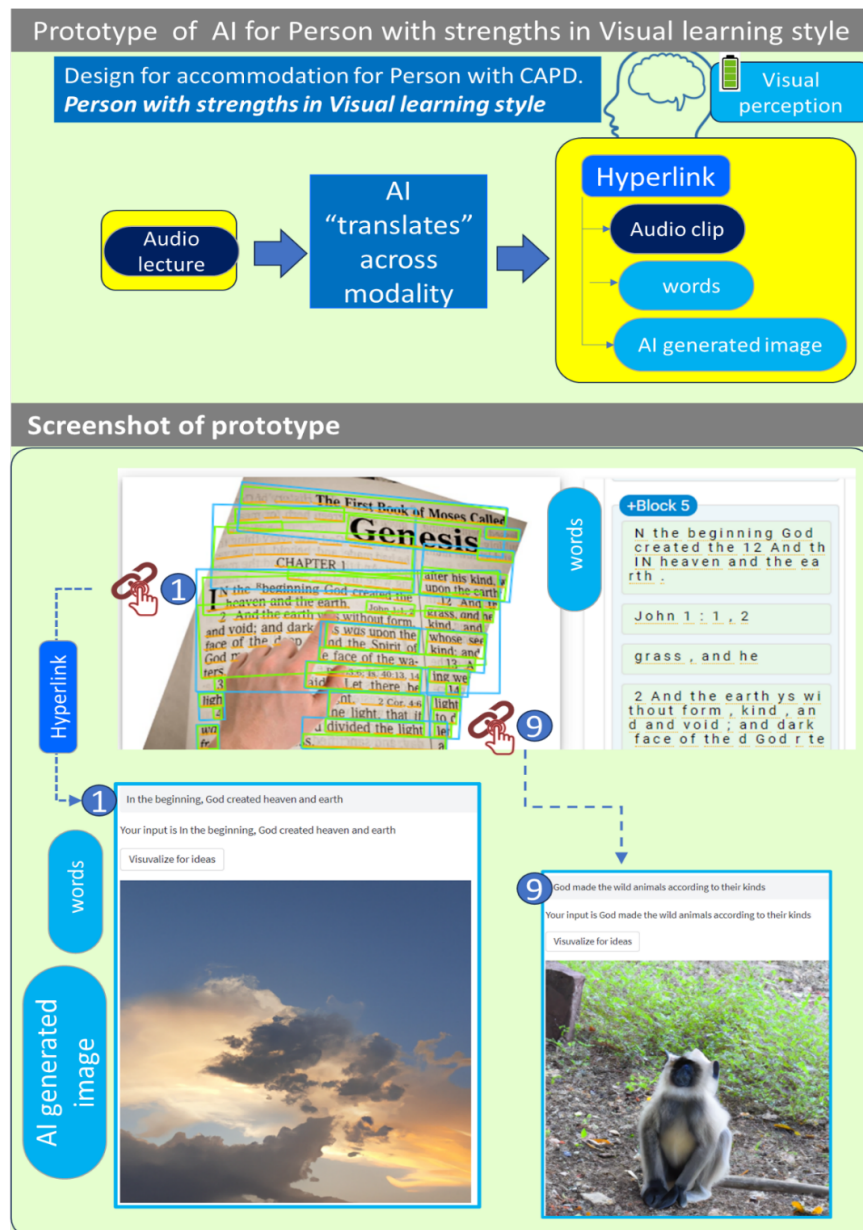


Figure 6: Generative Multimodal output to enable person with CAPD

visual or auditory perception. The paper explores a set of solutions to address many unique challenges [2] faced by individuals with Learning Disabilities by creatively mapping the challenges into multimodal tasks that can be executed by instructing a pretrained large Multimodal Foundation model [5-9] or by Multimodal latent space arithmetic [10]. With working demonstrations, the paper demonstrates that the challenges of persons with LD can be solved easily. Both Table 1 & 2 articulates the proposed solution approaches to various challenges, and the online project website demonstrates the set of solutions to enable persons with difficulty in visual/auditory

perception. The solution demo videos are presented online at <https://sites.google.com/view/multimodallearningdisability>.

1.5 Related work

The progress in Multimodal Generative AI [7] in the very recent time has been significant. Given rapid research advancements, the applications of Multimodal Generative Foundation models [7, 11, 12] for persons with Learning Disability [3] are yet to be fully explored. There are some key new shifts/trends to be taken note of. These new trends have opened up a significant research gap in its application for persons with learning disability. These

new trends have provided significant opportunities for researchers to contribute. The era of large scale Deep Learning Foundation models started with unimodal (sentence in, sentence out) [8], such as large language models (LLM) [21]. In contrast, the more recent multimodal LLMs support multimodal prompts [13] as inputs, and multimodal outputs [11, 20]. The relevance of unimodal vs multimodal models to address the challenges of LD significantly vary, given the challenges for VPD/CAPD often require transforming information across visual and auditory modality. Such cross-modal linkages and multimodal embeddings are discussed in [10]. There is a shift in trends in the field typically in the size & scale of neural network parameter size, compute power, and dataset size, giving rise to the era of large scale Deep Learning Foundation models [21]. Also trending is a shift from Transfer Learning to Few shot Learning [8], a big step in the number of training records needed to fine tune a model for a custom task. Another shift is in the number of tasks that can be performed by a Deep Learning model. While Deep Learning model architectures can be designed to support any type of tasks by configuring the model architecture before the start of training, Foundation models enable the models to be tweaked to perform any custom tasks simply by changing the instructions or prompts [7]. So the ability to learn from a handful of examples (Few shot learning [8] or One shot Learning [9]), and the ability for the model to perform any new or unseen task by prompt design provide significant shift in the paradigm. In short, the opportunity to conceptualize novel solutions for the VPD/CAPD is opening up given these recent shifts in the field of AI:- Multimodal Incontext tuning, Few shot learning, Multimodal prompts, Pretrained Foundation models, Multimodal embeddings, Multimodal Transformer architecture based Multimodal LLMs, Vision Language Models. By applying these new advances in AI to address the challenges of persons with APD, AI can be employed to generate accommodations [2, 28] such as Visual cues [29], “Look and then Listen” [30], Clear Language [31], Clear Speech [32].

With the scalable architecture of Transformers, there is significant progress in Vision Language modelling [5] and Multimodal Foundation models [7, 10], and its applications. The Transformers based joint learning of Visual and Language modality was powerfully exhibited in VLM (Vision Language Modelling) models such as ViLT and GIT [5, 6] during 2020-2022. GIT achieved human level parity. The progress in Multimodal Foundation LLMs in 2022-2023 is really accelerating. The application of some of the recent AI models for addressing LD challenges is a research gap. This paper not only designs with some of these latest AI models, but also prototypes to create lab scale demo using some the recent AI models such as OTTER [7], ImageBind [10], DALL-E [9], GPT3.5 [9], GIT [6], KOSMOS-2 [11]. Creating applications using these upcoming AI models is made easy by frameworks like Langchain [22] and Microsoft Sementic Kernel [23]. With the fast pace of AI research in recent few months, the opportunity for researchers to apply this progress for the benefit of person with disability is enormous.

2 DESIGN & DEVELOPMENT OF GENERATIVE MULTIMODAL ASSISITVE AI AGENTS (MAA)

2.1 Design of an AI system that dynamically composes many AI models that adapt to the user’s strengths & needs

The design principle is to create an approach that let’s person with Learning disability (LD) to overcome the challenge by creating a technology that assists based on the strengths of each individual. Given each individual have unique strengths or preferences in learning styles (auditory learning style or visual learning style), the design of Assistive Technology is personalized to the strengths of each individual. Initially the system identifies the strengths of the individual by creatively posing a simple set of challenges that combine different propositions of visual and auditory information, and assesses the strengths. The identification of the strength of the individual is also enabled by the creative use of multimodal latent space arithmetic, where visual and textual information is presented to the user by creative use of multimodal embedding space (refer to the online live demonstration at the project web page). Once the strength of the individual is identified, the system plans an appropriate intervention mechanism. For a person with VPD, the information from the environment will be translated into a format so as to make it easy for comprehension by the individual. When a person with VPD is accessing an e-learning module with a rich visual medium of communication, the AT attempts to translate the visual medium into a non-visual format. For a person with CAPD, the information from the environment will be translated into an appropriate output format. Thus, the output of the Multimodal Assistive AI agents (MAA) is tuned to suit the strengths of the individual. User appropriate intervention/ accommodation [2, 26, 28] is achieved by creatively tapping into the advances in Multimodal/Generative AI (more details is later articulated in Table 1 & 2). Based on the user’s learning preferences between audio learning style or visual learning style, the output of MAA may be synthesized speech or AI generated visual. Such dynamic adaption to the user’s strength is achieved based on strength identification of the user & human feedback to personalize the output modality to the desired format. So here the design principle is to offer an appropriate accommodation to suit the individual’s strengths/preferences. The accommodation to user accessible modality is illustrated in Fig 1 and Fig 2 for VPD and CAPD.

The MAA can be a smartphone app or a laptop software, thus enabling the person with LD to interact with teacher/co-worker, and also enabling them in digital world. The MAA technology is practically feasible to deploy at scale, and be personalized to thousands of persons with LD. This is because of the advances in Generative/Multimodal AI and also due to the availability of frameworks like Microsoft Semantic Kernel [23] and langchain [24]. Once the strengths of user are identified, Langchain agents allow dynamic composition of multiple AI models into a pipeline, thus providing an implementation framework for dynamic selection of AI models and chaining the intelligences together into an intelligent chain of AI models as required by the user & environment. This architecture is generalized in Fig 8. The multimodal interactions can be in near real-time (like when listening to a teacher’s lecture) or

Table 1: Solving the challenges of persons with VPD

	Challenges for persons with VPD (Visual processing disorder)	How to address the challenge?	Relevant AI approach	Demo
1	Person experiences hindered ability to make sense of information taken in through the eyes [2, 26]	AI powered Accommodation: Transform visual info into an Interactive oral Question Answering human computer interaction	Visual Question Answering with ViLT, Multimodal LLM., OTTER	Yes
2	Identify the strengths of the individual as either visual or auditory learner [27]	Identify Strengths: Assessment of user's strengths (visual vs auditory) and dynamically adjust the proposition of audio/visual mix based on continuous user's feedback	Multimodal Latent space arithmetic, Human-in-the-Feedback loop	Yes
3	Spatial difficulty (Difficulty perceiving spatial depth & directionality) [2, 26]	Joint modelling of depth images & text: Represent Depth images in the form of language using Multimodal embeddings	Emergent alignment between depth and text via IMAGEBIND	Yes
4	Difficulty in understanding Whole/part relationships [2, 26]	Instruct an AI model: Multimodal In-Context Instruction Tuning	One shot learning in Multimodal LLM, OTTER	Yes
5	Visual reasoning difficulty [26]	Instruct an AI model using an example: In-Context Instruction Tuning of Multimodal LLMs & Few shot learning	Instruction Tuning & Few shot learning in Multimodal LLM, OTTER	Yes

Table 2: Solving the challenges of persons with CAPD

	Challenges for persons with APD (Auditory Processing Disorder)	How to address the challenge?	Relevant AI approach	Demo
1	Person experiences hindered ability to make sense of information taken in through the ears [2, 28]	AI generated visual cues or multimodal cues: Generative AI transforms audio info into a visual storyline or a multimodal format	Employ Generative AI models to generate multimodal/visual cues	Yes
2	Background noise in the audio lecture, and the speaker is talking fast	AI generated Clear Speech: AI speaks at a reduced rate to improve signal perception [30]	AI speaks at a reduced rate by using Speech recognition & then Speech Synthesis	Yes
3	Audio lecture consists of long complex sentences	AI generated Clear Language : LLMs rephrase complex sentences into simpler sentences. (easy for comprehension) [31]	Instructing a Large Language model with few examples to rephrase to simple sentences	Yes
4	The person experiences difficulty in auditory memory	AI retrieved visuals: Help remember by retrieval of associated Visuals	Cross-modal retrieval of personal photos that is associated with an audio using Vector search of Multimodal embeddings	Yes
5	Difficulty in multi-sensory perception	AI generated Cross-modal hyperlinks: Enable the person to first Look at the information, and then listen the same info "Look and then Listen" [30]	AI generates Multimodal document with Hyperlinks across visuals & words/audio	Yes

time lapsed (e-learning session or presentation at work), where the AI translates content. Thus, MAA allows a dynamic composition of the pipeline of different AI models to build an intelligent assistant that adapts to the dynamic needs of each individual.

2.2 Solving the challenges by creative use of Multimodal/Generative AI models

One of the key contributions of this paper is solving the many challenges for persons with LD. The challenges are addressed by various AI approaches as articulated in Table 1 & 2, and

then further elaborated in the next two sections. The feasibility of the implementation of these approaches are demonstrated by experimental lab scale prototype development, and then recorded as videos and published in the project URL at <https://sites.google.com/view/multimodallearningdisability/>

2.3 MAA for enabling Persons with strengths in Auditory learning style, and difficulty in visual perception

Verbal user interaction with visual data: As illustrated in Fig 1 & Table 1, MAA transforms visual input into an interactive multi-modal conversation. We developed a lab scale prototype implementation to address the challenges of persons with VPD. The design enables the child to ask questions about the image with Visual Question Answering (VQA). The user's question and input visual is encoded by the Vision Language AI models [5] to

produce an answer. We implemented a prototype with two different AI models, ViLT [5] and GIT [6]. Either ViLT or GIT model can be composed based on requirements on user perceived latency vs accuracy tradeoffs. The screenshot of the prototype is shown in Fig 3. The online demo of the prototype implementation with both ViLT and GIT will be available at the project website at <https://sites.google.com/view/multimodallearningdisability>.

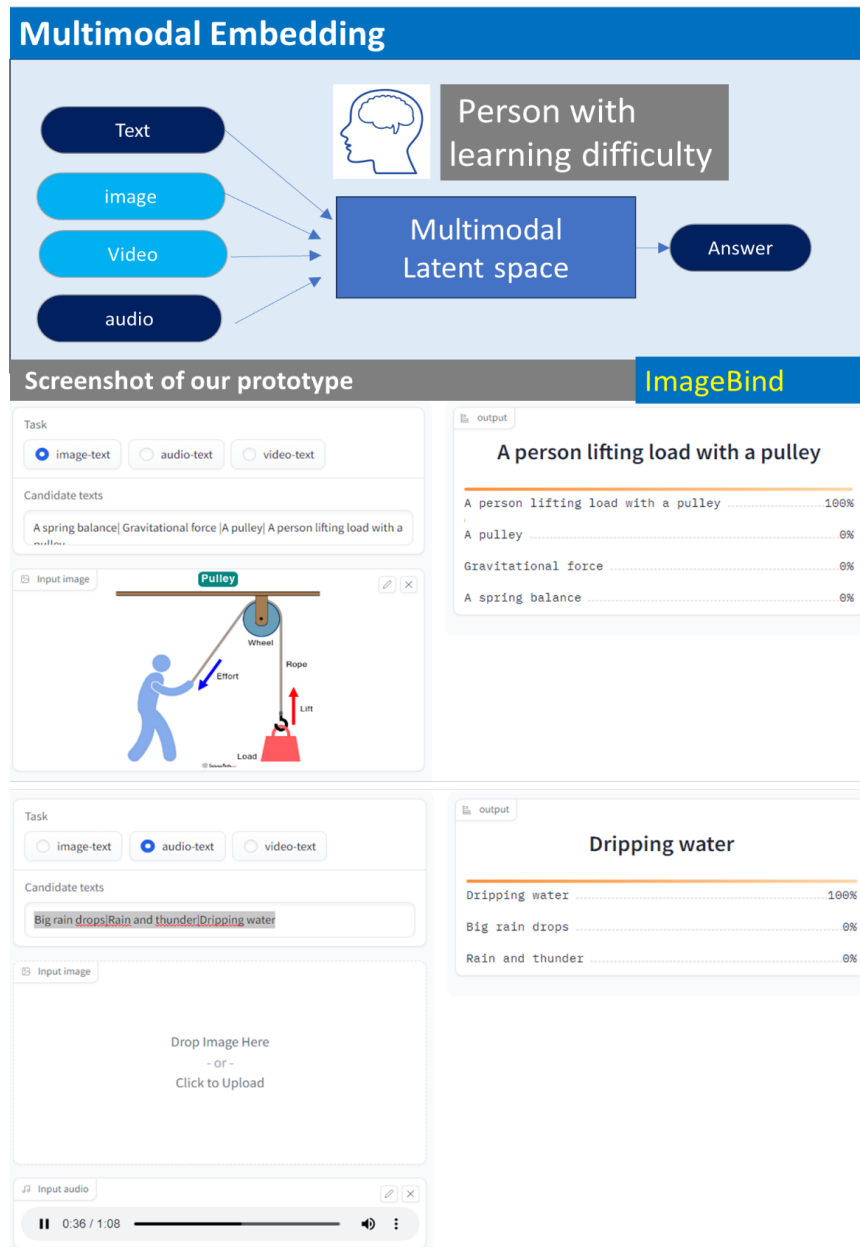


Figure 7: Screenshot of our ImageBind prototype

2.4 MAA for enabling Persons with strengths in Visual learning style, and difficulty in auditory perception

MAA provides accommodations as articulated in Table 2 and Fig 2. When a person with CAPD faces a scenario of any spoken instruction or audio lecture, the audio instructions are transformed by MAA into an AI generated hyperlinked multimodal document. This output multimodal content is a mix of AI generated content combining visual cues, words and audio clips. This enables “Look and then Listen” [29, 30] usage model, where the individual can first look at the visual cues and click on the hyperlinks to hear the corresponding word as an audio clip. Thus, instead of listening to an audio lecture, the individual can interact with the content with a new Human computer interaction model. A video demonstration of this AI powered “Look and then Listen” [30] accommodation is presented at the project webpage at <https://sites.google.com/view/multimodallearningdisability>. As noticed in the video, Large Language models (LLM) such as GPT3.5 is employed to rephrase complex spoken sentences into simpler phrases, and Generative image generation models generate visual cues, and these visual cues and simpler phrases are hyperlinked. Thus, the user can look at the visual cues and the phrases, and click on the phrases to hear a corresponding audio clip to have a “Look and then Listen” [30] Human Computer Interaction experience, thus enabling individuals with difficulty in multi-sensory integration to process sequentially, first the visual data and then the audio data. This AI generated multimodal document has hyperlinks across elements such as “AI generated Clear Speech/Language” & “AI generated Visual cues” and “AI retrieved visual memory records”. The Microsoft Immersive Reader [33] is employed to enable “Look and Listen”. “Clear Speech” [32] is enabled simply by Speech Recognition followed by Speech Synthesis, where the engine is set to speak at a slower speaking rate, thus enabling the individual to listen. Thus, AI generated “Clear speech” enables improved signal perception, when the speaker speaks fast in an environment with background noises. “Clear Language” [31] is about rephrasing to improve comprehension, where LLMs are employed. If the word “beverage” is not yet in the child’s vocabulary, the system uses the word “drink”. Thus, by instructing the LLMs to rephrase to simpler known phrases & use word choices from a limited vocabulary, AI generated “Clear Language” is enabled. The visual cues such as images & image frames/videos are generated by AI to communicate the storyline. When the individual has difficulty in auditory memory, we employ cross-model image retrieval to retrieve photos from a personal album that correspond to a sound. Here multimodal embeddings that link images and text, followed by vector search is performed to retrieve “AI retrieved visual memory records”. The visuals in the multimodal document can be AI generated visuals or AI retrieved visuals. The video demonstrations of the solutions can be viewed at <https://sites.google.com/view/multimodallearningdisability>.

3 FUTURE DIRECTIONS & CONCLUSIONS

Future directions: As UN Convention of Rights of Persons with Disability protects the rights of persons of persons such as the right to contribute as equals in workplace, it is imperative to research and develop innovative approaches to augment human capability. In

the era of AI based copilot to unlock human potential, any research on augmenting the capability of persons with learning disability can lead to a substantial impact in the lives of people. Given the fast pace of progress in Generative/Multimodal AI and related frameworks to compose a set of AI models into a dynamic pipeline that combines multiple intelligences, the ability to create very intelligent multimodal systems is becoming real. This future possibility of the dynamic composition of AI models to create a dynamic intelligence is visualized in Fig 8. This allows the proposed multimodal design to be best suited for each individual strengths, as each child/person is unique.

Summary/Conclusions:

With the rapid advances in the field of Generative/Multimodal Foundation models in a very recent times, the opportunity to employ these advances for the benefit of Persons with Disability is dawning. Given this wide opportunity to apply the recent advances in AI for the benefit of persons with disability, this paper attempted to map the challenges of persons with learning disorders into problem statements in Generative/Multimodal AI. We designed an approach to Generative Multimodal Assistive AI agents (MAA) as a way to assist and enable persons with Learning disability. Based on the difficulty faced by the person/child, MAA dynamically enables the person with learning difficulty to be successful in both school and at work. We designed and developed MAA for two common types of learning disorders, visual and auditory disorders. The paper specifically designed solutions for various challenges in VPD (Visual Processing Disorder) and CAPD (Central Auditory Processing Disorder). This paper provides a state-of-the-art design of Multimodal Assistive Technology by bringing in the latest research progress in the field of AI to enable persons with VPD/CAPD to succeed.

The paper mapped the challenges of VPD/CADP into a design approach, and prototyped lab scale demos to check the feasibility of translating these ideas into reality. Based on the design approach of multimodal assistive agents, and a dynamic composition of a set of AI models, the paper seeds early thoughts of inspiration. The challenge of persons with various types of learning disability was solved by creative application of AI models. The design and prototype explored how to apply some of the recent AI models to enable persons with Visual or Auditory processing disorders. Inspired by unimodal ChatGPT/GPT3.5 conversational systems, this paper designed and developed powerful multimodal conversation AI agents to serve the needs of persons with neurological conditions. We designed & developed Multimodal Assistive AI agents for enabling persons with difficulty in Perception in visual/spoken modalities. This design can be extended to mainstream users, as this multimodal person assistant can adapt to the learning style of the individual such as visual learning style or auditory learning style. These prototypes lead the path for future development. When the individual faces difficulty in interpreting complex visual inputs, MAA employs Interactive Multimodal Translation to bring user accessible content. When the individual faces difficulty in visual discrimination, MAA demonstrated Verbal Interaction with visual content. When the individual faces difficulty in visual memory or difficulty in learning math, MAA demonstrated conversational interaction & Visual Reasoning over multiple images. When the person faces difficulty to understand orally spoken directions, MAA

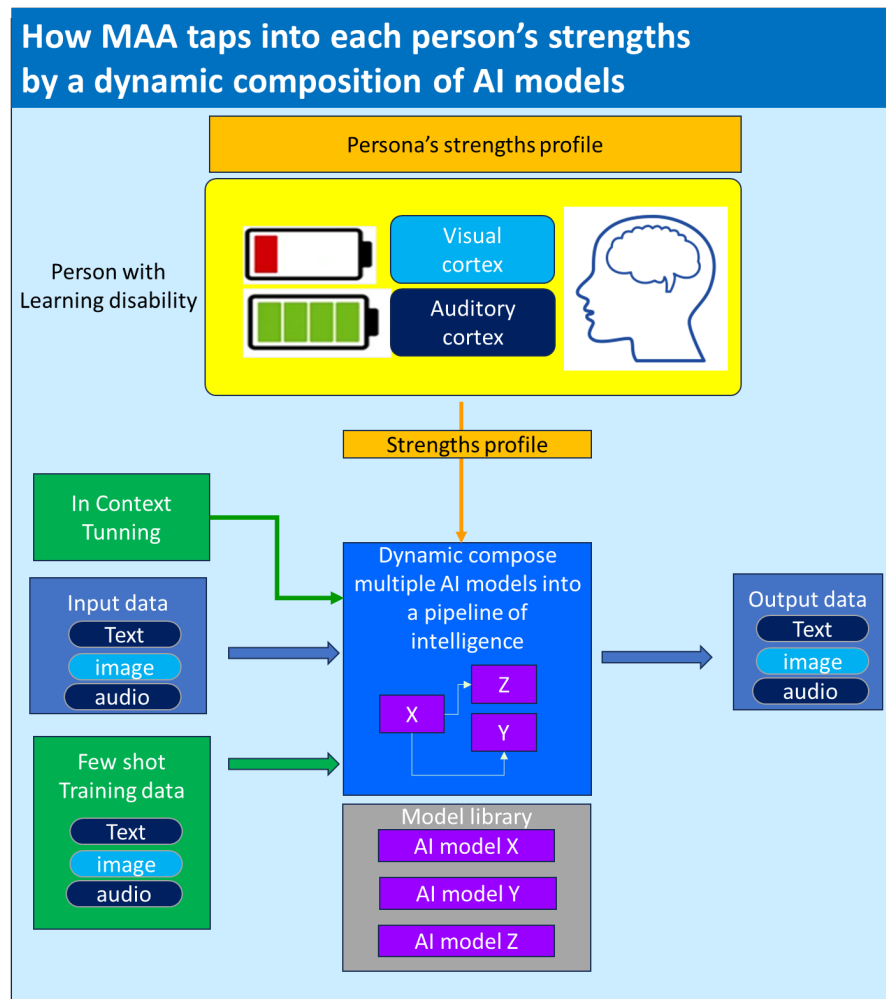


Figure 8: Future directions for future research

demonstrated the generation of hyperlinked set AI generated visual cues & clear language/speech. The paper was focused on enabling persons with disabilities by tapping into the latest advances in AI such as Multimodal embedding, Multimodal Generative models, Multimodal In-Context learning, Few shot learning. In short, the paper seeded Multimodal Generative AI as a design approach to solve the many challenges for persons with learning disability, and demonstrated the feasibility of implementation by implementing lab scale working prototypes. The videos/demonstrations of how MAA (Multimodal Assistive Generative AI) framework addressed the many challenges are posted on the project webpage at the URL, <https://sites.google.com/view/multimodallearningdisability>.

REFERENCES

- [1] National Center for Education Statistics. National Center for Education Statistics. (2023). Students With Disabilities. nces.ed.gov/programs/coe/pdf/2023/cgg_508.pdf
- [2] National Center for Learning Disabilities (NCLD). Visual and Auditory Processing Disorders | LD OnLine. www.ldonline.org/ld-topics/processing-deficits/visual-and-auditory-processing-disorders
- [3] Learning Disabilities Association of America. Learning Disabilities - Getting Help. <https://ldaamerica.org/support/new-to-ld/>
- [4] Horowitz, S. H., Rawe, J., & Whittaker, M. C. *The State of Learning Disabilities*. <https://www.nclld.org/Research/State-of-Learning-Disabilities/Executive-Summary>. National Center for Learning Disabilities.; 2017.
- [5] Kim W, Son B, Kim I. ViLT: Vision-and-Language Transformer Without Convolution or Region Supervision. *arXiv:210203334 [cs, stat]*. Published online June 10, 2021. <https://arxiv.org/abs/2102.03334>
- [6] Wang J, Yang Z, Hu X, et al. GIT: A Generative Image-to-text Transformer for Vision and Language. *arXiv:220514100 [cs]*. Published online December 15, 2022. <https://arxiv.org/abs/2205.14100>
- [7] Li, B., Zhang, Y., Chen, L., Wang, J., Yang, J., & Liu, Z. (2023). OTTER: A multimodal model with in-context instruction tuning. *arXiv:2305.03726* <https://doi.org/10.48550/arXiv.2305.03726>
- [8] Brown TB, Mann B, Ryder N, et al. Language Models are Few-Shot Learners. *arxiv.org*. Published online May 28, 2020. <https://arxiv.org/abs/2005.14165>
- [9] Ramesh A, Pavlov M, Goh G, et al. Zero-Shot Text-to-Image Generation. *arXiv:210212092 [cs]*. Published online February 26, 2021. <https://arxiv.org/abs/2102.12092>
- [10] Girdhar R, El-Nouby A, Liu Z, et al. ImageBind: One Embedding Space To Bind Them All. *arXiv.org*. doi:<https://doi.org/10.48550/arXiv.2305.05665>
- [11] Peng Z, Wang W, Dong L, et al. Kosmos-2: Grounding Multimodal Large Language Models to the World. *arXiv.org*. doi:<https://doi.org/10.48550/arXiv.2306.14824>
- [12] Tang Z, Yang Z, Zhu C, Zeng M, Bansal M. Any-to-Any Generation via Composable Diffusion. *arXiv.org*. doi:<https://doi.org/10.48550/arXiv.2305.11846>
- [13] Liu V. Beyond Text-to-Image: Multimodal Prompts to Explore Generative AI. Published online April 19, 2023. doi:<https://doi.org/10.1145/3544549.3577043>

- [14] Gozalo-Brizuela R, Garrido-Merchán EC. A survey of Generative AI Applications. arXiv.org. doi:<https://doi.org/10.48550/arXiv.2306.02781>
- [15] Olga A, Tzirides A, Saini G, *et al.* Implications and Applications for Education. Published online 2023. doi:<https://doi.org/10.48550/arXiv.2305.07605>
- [16] Assistive Technology and Universal Design: A Toolkit for Interagency Collaboration | Interagency Committee on Disability Research. icdr.acl.gov. Accessed July 31, 2023. <https://icdr.acl.gov/resources/reports/assistive-technology-and-universal-design-toolkit-interagency-collaboration>
- [17] Yuasa R, Tamura A, Kajiwaru T, Ninomiya T, Kato T. Multimodal Neural Machine Translation Using Synthetic Images Transformed by Latent Diffusion Model. ACLWeb. Published July 1, 2023. Accessed July 31, 2023. <https://aclanthology.org/2023.acl-srw.12>
- [18] Kuo W, Piergiovanni AJ, Kim D, *et al.* MaMMUT: A Simple Architecture for Joint Learning for MultiModal Tasks. arXiv.org. doi:<https://doi.org/10.48550/arXiv.2303.16839>
- [19] Nir Kshetri. ChatGPT in Developing Economies. 2023;25(2):16-19. doi:<https://doi.org/10.1109/mitp.2023.3254639>
- [20] Sundar A, Heck L. Multimodal Conversational AI: A Survey of Datasets and Approaches. Published online January 1, 2022. doi:<https://doi.org/10.18653/v1/2022.nlp4convai-1.12>
- [21] Jeon J, Lee S. Large language models in education: A focus on the complementary relationship between human teachers and ChatGPT. *Education and Information Technologies*. Published online May 2, 2023. doi:<https://doi.org/10.1007/s10639-023-11834-1>
- [22] Lobentanzer S, Saez-Rodriguez J. A Platform for the Biomedical Application of Large Language Models. arXiv.org. doi:<https://doi.org/10.48550/arXiv.2305.06488>
- [23] Semantic Kernel. GitHub. Published August 1, 2023. Accessed August 1, 2023. <https://github.com/microsoft/semantic-kernel>
- [24] Langchain, GitHub. Published August 1, 2023. Accessed August 1, 2023. <https://github.com/langchain-ai/langchain>
- [25] Li B, Zhang Y, Chen L, *et al.* MIMIC-IT: Multi-Modal In-Context Instruction Tuning. arXiv.org. <https://doi.org/10.48550/arXiv.2306.05425>
- [26] Valerie L. Frazer, Visual Skills for Reading and Learning: Vision Perception Webinar, <https://www.newhorizonsvisiontherapy.com/category/vision-and-learning/>
- [27] Strengths of Students with Learning Disabilities, The National Center for Learning Disabilities, [https://www.youtube.com/watch?v=\\$CYHzJGTA6KM](https://www.youtube.com/watch?v=$CYHzJGTA6KM)
- [28] Deborah Ross-Swain, Donna S. Geffner. Auditory Processing Disorders: Assessment, Management, and Treatment, Plural Publishing, ISBN 1944883428
- [29] Helfer, Karen S. "Auditory and auditory-visual perception of clear and conversational speech." *Journal of Speech, Language, and Hearing Research* 40.2 (1997): 432-443.
- [30] Managing Children's central auditory processing deficiencies in the real world. what teachers and parents want to know. Ferre, 2002, *Seminars in Hearing*, 23, 319-326
- [31] Ferre, Jeanane M. "Understanding intervention for (C) APD: as easy as ABC." *The ASHA Leader* 12.10 (2007): 20-22.
- [32] Picheny, Michael A., Nathaniel I. Durlach, and Louis D. Braida. "Speaking clearly for the hard of hearing I: Intelligibility differences between clear and conversational speech." *Journal of Speech, Language, and Hearing Research* 28.1 (1985): 96-103.
- [33] Jeanneret Medina, Maximiliano, *et al.* "It Deserves to Be Further Developed: A Study of Mainstream Web Interface Adaptability for People with Low Vision." CHI Conference on Human Factors in Computing Systems Extended Abstracts. 2022.